

Two call centres on one system, calling from the screen on 1 October

What was built on 26–28 September 2026: a cross-tenant leak closed, stepwise onboarding that can give an organisation ConnectDesk alone, ConnectDesk working Fastdial's own CallBox database, the Phase 2 floor on the server, and the calling architecture. That architecture covers click-to-call through the PBX today and a WebRTC phone, with conference, transfer, listen, whisper and barge, once the media edge is installed. Since the 27th it also answers the gap the PBX's records exposed: customers who ring Fastdial reach voicemail. There is now a missed-calls inbox, a dialler that returns their calls first, and an inbound line that rings Ready agents. Adherence, call recordings, AI call review, quality sampling, Erlang C staffing and a wallboard for the floor's TV were added too. So were per-lender client reports that finally say whether promises were kept, a contact policy that stops the dialler over-calling anyone, and, for each agent, their own day and a call guide. It closes with the go-live list for both companies.

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Prepared	For	Pilot starts	6ba07b0 → b4926ba

01 Where it stands

PIECE	STATE	WHAT THAT MEANS
Cross-tenant leak in ConnectDesk, PeopleHub, Ledgerly	Fixed · live	Every CollectBox read now belongs to the signed-in organisation. Before this, any org holding those systems saw Micromart's book.
Onboarding v2 (company → location → systems → admin → brand)	Live	/onboard . ConnectDesk-only is a valid choice; each system answers on <system>.servicesuitecloud.com/<slug> the moment the org exists.
Fastdial as a ConnectDesk-only organisation	Live	connectdesk.servicesuitecloud.com/fastdial.lms.../fastdial redirects there. Supervisor and Agent roles exist.
ConnectDesk on Fastdial's CallBox database	Live · writes on	Floor, campaigns and builder, work lists, case files, outcomes, promises, callbacks, contracts, agents and phone floor, all on FastDialCallBoxMaster.
Phase 2: the floor on the server	Live	Real seats, logins, lateness, breaks, overruns, pauses, bucket moves and the floor log, for both companies. A live stream feeds every screen.
Adherence report	Live	/desk/adherence : logins, lateness, breaks, overruns and pauses against the roster, with the PBX's records as a second witness. CSV for payroll.

Missed calls, and returning them first	Live	/desk/inbox : every customer who rang and reached voicemail, matched to their account, with whether anyone called back.
The dialler (progressive and preview)	Built - off until switched on	Hands a Ready seat its next customer, missed callers first. Dialler settings → <i>Hand out calls automatically</i> .
The inbound line (the floor's own ACD)	Built - needs the edge	The edge registers as 716 and rings Ready agents, the account's owner first. Needs the edge box and 716's password.
Recordings in the case file	Built - test from prod	Plays the PBX's own recording of each call. Same reachability as click-to-call (\$08 step 1).
AI call review	Built - off until agreed	Transcript, summary, promise, flags and QA from a recording, one call at a time. Off until each company agrees to it.
Staffing (Erlang C)	Live	/desk/staffing : the week hour by hour, and the agents the inbound line needs at a chosen service level.
Wallboard	Live	/desk/wallboard : the floor on the TV — today's numbers, every seat live, customers ringing in.
Quality	Live	/desk/quality : recorded calls sampled per agent, playable in the row, with AI scores where reviewed.
Client reports	Live	/desk/reports : one lender's book over any range: work, promises kept or broken, money collected, by day, agent and contract. Prints to PDF, and each table exports as CSV.
Promise verdicts	Live	The promise board and case file mark each promise kept, part-paid or broken from the payments on the account. CallBox itself never records this.
For agents: My day and the call guide	Live	An agent's home shows their calls, promises, promises due, broken promises to follow up and money credited this month. Every case file carries a call guide worded for the customer on screen.
Campaign results	Live	Each campaign card shows accounts reached, promises taken and money paid since it opened.
Contact policy	Built - acts once the dialler is on	The dialler holds back anyone called or spoken to too often this week, who has just paid, who promised a later date, or whose last outcome is on the do-not-dial list.
Calling: desk phone (PBX click-to-call)	Built - test from prod	Call buttons ring the agent's extension through the Yeastar API, then the customer. Proven against the PBX from this workstation; the production test is one click (\$08).
Calling: browser phone (WebRTC)	Built - needs the edge	The phone and the edge configuration are built. It goes live when the edge box is installed (≈45 min, runbook in <code>bridge/edge</code>).
Micromart calling	Needs their PBX	The floor is live. Calls stay simulated until Micromart's PBX type, API and SIP details are in Settings → Connections.

02 The leak, fixed first

Every ConnectDesk page, both desk APIs, PeopleHub and Ledgerly called `collectBoxOrg("micromart")` — a constant — whoever was signed in. Every lender's `Org.systems` was `null`, which means "all systems", and Fastdial's Org Admin holds every right. So Fastdial's administrator, and the staff of Mular, Axe, BuySimu and Techcrast, could open `/desk`, `/people` or `/books` and see Micromart's collections book, staff directory and journal. The desk's write path would also have recorded their outcomes into Micromart's CollectBox, under a fallback agent.

`lib/collectbox/tenant.ts` now resolves CollectBox from the verified session. Only an organisation that owns a CollectBox floor gets one, and everyone else sees the pages' existing "not connected" state. It shipped on its own as `6ba07b0` and deployed successfully.

03 Onboarding, and an address for every system

The wizard is six steps in a landscape split: a rail with the steps and a live identity card on the left, the step on the right.

- **Company:** trading and registered name, what the company does (lender, MFB, SACCO, bank, call centre / BPO), KRA PIN (checked for shape: letter, nine digits, letter), registration number, email, phone, website.
- **Location:** address, town, county (all 47) and a map pin, with latitude and longitude. It uses the Google map where a key is set, and OpenStreetMap otherwise.
- **Systems:** any combination of the seven, with presets ("Call centre" = ConnectDesk alone). The organisation's address is checked live as it is typed, with every system's resulting link shown.
- **Administrator, Branding** (logo and colours on every door and email), then **Review**.

`POST /api/orgs` stores the profile, an explicit `Org.systems` list (never null) and starter roles that follow the systems. A ConnectDesk organisation gets *Call-centre Supervisor* and *Agent*, using two new rights, `desk.supervise` and `desk.dial`.

No deployment is involved in creating an organisation's address. The door at `/<slug>` answers on every host and wears the org's logo. A door the org does not hold sends visitors to one it does: Fastdial typing `lms.../fastdial` lands on ConnectDesk. `/platform` has an *Onboard an organisation* button.

04 One ConnectDesk, three kinds of floor

ConnectDesk asks, for the signed-in organisation, where its book lives (`lib/desk/source.ts`), and renders the matching screens and menu:

KIND	EXAMPLE	BOOK	BUCKETS
collectbox	Micromart	CollectBox on its ServiceSuite SQL Server, through the bridge	The seven arrears bands
callbox	Fastdial	CallBox Postgres (vault: <code>CALLCENTRE</code>)	Open campaigns
none	A new call centre	Nothing yet. The floor, settings and phone work; the book waits for Settings → Connections	—

Fastdial's CallBox, as it actually works

Read from the code (`C:\GIT\CallBox`) and the live schema, read-only, on 26 September:

- **companies** are the lenders Fastdial collects for: 15, including KCB Digital and Conventional, 4G, Zenka, Pi Capital (Bayes), Kingdom Bank and Mular Credit.
- **contracts** (50) are uploaded batches. **contractdata** (151,789) has one debtor per row, with `olb`, `arrears`, `assignedagent`, `calls_made`, `actionstatus` (the last outcome) and `campaignid`.
- A **campaign** is a filter over one contract for a set of agents: balance range, arrears range, calls made, last outcomes, worked before or not, paid something or not, and last-worked dates. Creating it *locks* every matching row that is in the pool and assigned to those agents. Saving an outcome puts the row back in the pool (`campaignid = 0`, which means available, not closed), and the campaign closes itself when empty.
- **Outcomes** come from `paymentresponse` (21 of them). Some schedule a promise (a `promisedtopay` row) and some a callback (`task_scheduler`).
- The PBX's own event receiver on `213.148.17.228:8260` writes `callrings` and `calldr` live (3,229 CDRs in the last 24 h when checked), keyed by the Yeastar call id.

Same statements, side by side

ConnectDesk writes with the statements CallBox itself runs, so the two can work the same floor through the pilot and each is true of the other's work. Every write path ran against the live schema inside a transaction that was then rolled back. The record was unchanged afterwards and no campaign row was added.

CONNECTDESK	MIRRORS CALLBOX	PROVEN
<code>logCallStart</code>	<code>HomeController.SaveCallLog</code>	Call-log row with the PBX call id
<code>saveDisposition</code> (campaign)	<code>CallsController.PostCallResponse</code>	Outcome, promise, callback; row back to the pool; campaign auto-close
<code>saveDisposition</code> (one-off)	<code>CallsController.PostSearchResponse</code>	The customer stays in their campaign
<code>createCampaign</code>	<code>CallsController.SaveCallCampaign</code>	Locked 908 rows; the live preview counted 908
<code>closeCampaign</code>	<code>CallsController.CloseCampaign</code>	Released 1,030 rows (rolled back)

Two deliberate improvements: an outcome's rules (a promise needs an amount and a date, a callback needs a time) are enforced on the server, and the follow-up task is written inside the transaction, where CallBox writes it on a separate connection. CallBox's timestamps are Nairobi wall time; ConnectDesk reads them as +03:00, so nothing shifts three hours on Vercel.

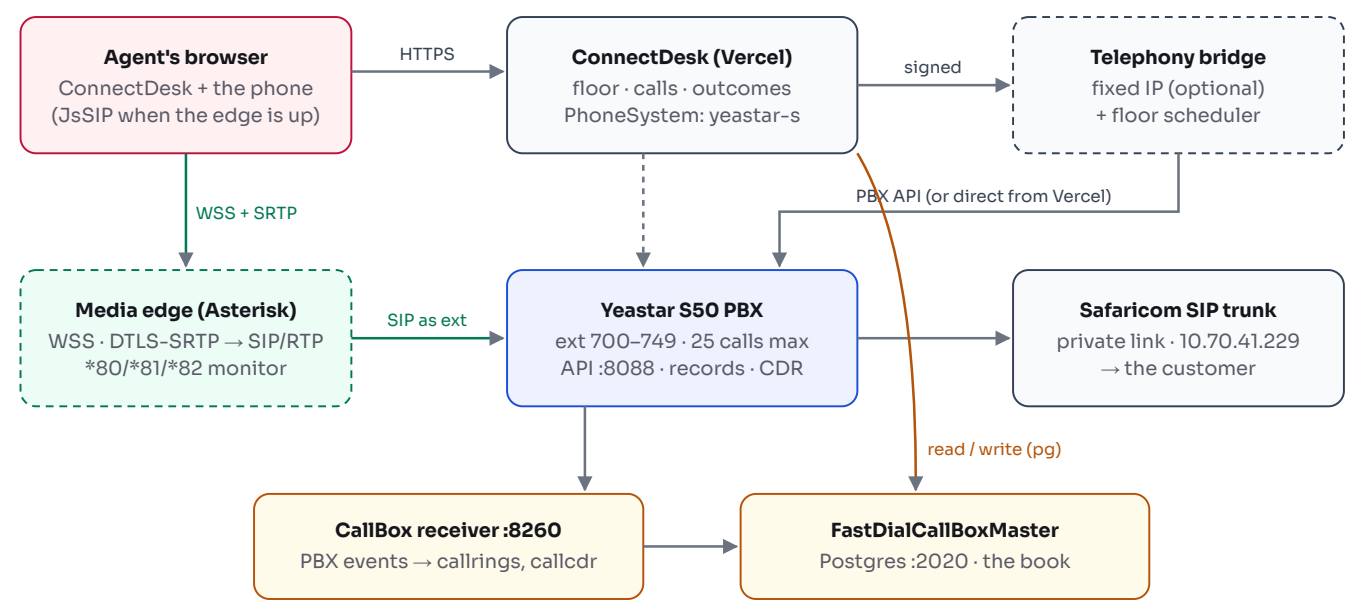
05 Phase 2 — the floor on the server

The Phase 1 engine's pure rules are still the rules: `apply` for actions, plus a new `realTick` for time. The tick runs the schedule's 08:00 start and 17:30 stop on working days, the overrun alert, pause expiry and wrap-up to ready, with none of the simulated calls. They now run on the server (`lib/desk/floor/server.ts`) over:

TABLE	HOLDS
DeskFloor	The master switch, break types, schedule, dialling rules and bucket routing, and the once-a-day schedule flags
DeskSeat	One row per seat: login, extension, agent id in the org's own system, bucket, mode, and live state
DeskSeatSession	Every login and logout, with lateness in seconds (the adherence report's evidence)
DeskBreakLog	Every break with its allowance, overage and who recalled it
DeskFloorEvent	The floor log, append-only
DeskCall · DeskPbxSession	Every call placed through ConnectDesk, tied to the PBX call id; the shared PBX API token

- **Routes:** `GET/POST /api/desk/floor` (an agent opening ConnectDesk logs their seat in), `/api/desk/floor/stream` (server-sent events) and `/api/cron/desk-floor` (the scheduler for when no screen is open). Only changed rows are written, so actions on different seats never contend.
- **Permissions:** an agent can act only on their own seat. A supervisor can act on any seat, and in "View as" the log records the action as that agent's.
- **The roster seats itself** from the org's own system: 26 working agents from CollectBox for Micromart, and 31 from CallBox for Fastdial, with their campaigns as buckets. Importing agents (Settings → Seats) gives each one a login with the Agent role and a one-time password.
- **Verified end to end** on Fastdial: start floor → log in (late by the roster) → washroom → recall → move two seats to a campaign → pause → log out. It produced one session with lateness, one recalled break and nine log events. The test rows were then removed.

06 Phase 3 — how calls work



Solid: works on 1 October with nothing new installed. Dashed: the media edge (browser phone) and the bridge (only if the PBX's API refuses Vercel's addresses).

One pipeline, two ways to hear the call

Every call goes through `lib/desk/calls.ts`. It gets a `DeskCall` row, a `CallBox` call-log row carrying the PBX call id (exactly what `CallBox` writes), and a live state the phone and the case file follow: dialling → ringing → answered → ended.

DESK PHONE · DAY ONE

PBX click-to-call

The PBX rings the agent's own handset or softphone, auto-answers it and dials the customer: Yeastar `dial_outbound`, the mechanism `CallBox` already uses thousands of times a day. Hold, mute, transfer and hang-up go through the PBX API. The state comes from the PBX's live call list, cached two seconds per org (30 agents cost the PBX one request, not 30), and then from the CDR.

BROWSER PHONE · WITH THE EDGE

WebRTC as the extension

The browser registers through the edge as the agent's extension, and calls are SIP from the tab. The engine prefers G.711 so the edge never transcodes. The PBX still sees extension 715: its CDR, recording, outbound rights and monitor permissions are unchanged.

In-call capabilities

CAPABILITY	DESK PHONE	BROWSER PHONE	HOW
Click-to-call from any row, case file, promise or callback	Yes	Yes	One <code>desk:dial</code> event; the phone does the rest
Hold · resume · mute	Yes	Yes	PBX API / SIP re-INVITE
Keypad tones (IVRs, bank menus)	—	Yes	RFC 4733 from the browser
Blind transfer	Yes	Yes	PBX <code>calltransfer</code> / SIP REFER

Add someone · three-way conference	—	Yes	The second call is dialed while the first holds, then Merge mixes it in the tab: each party hears the agent and the other
Supervisor listen · whisper · barge	Yes	Yes	Seat drawer → PBX monitor API rings the supervisor's phone. On the edge, <code>*80/*81/*82+ext</code> works on any PBX
Outcome written to the call's own log row	Yes	Yes	The phone's wrap-up or the case file; the CallBox row carries the PBX call id
Recording	PBX	PBX + edge	The PBX records as today, and the edge can also record (a setting). ConnectDesk plays the PBX's recording in the case file and on the phone floor, and can have it reviewed by AI (\$07).
Customers ringing in	Yes	Yes	Through the inbound line on the edge (\$07): Ready agents are rung in turn, desk phones through the PBX, browser phones directly, and the incoming screen shows the caller's account.
Progressive and preview dialling	Yes	Yes	The dialler hands a Ready seat its next customer (\$07). Predictive (dialling ahead of agents) needs the edge's own queue: the Yeastar API rings the agent first

The PBX facts this rests on (review of 20 August)

Yeastar S50, firmware 30.15. It holds **50 of 50** extensions (700–749), so no new extension can be added without freeing one, and 23 placed no calls that day. Its ceiling is **25 concurrent calls** (peak measured: 18). The trunk is SAFARICOM-SIP over a private link, with FOX1 configured but idle. The S-Series has **no WebRTC**, which is why the edge exists. The API gives click-to-call, hold, mute, transfer, listen, whisper, barge and live call lists. ConnectDesk holds **one shared API token**, where CallBox logged in before every action (5,768 logins in one day). It also logs in with **exactly the report address CallBox's receiver uses**, because a Yeastar login re-points the PBX's live event feed.

07 Added on 27–28 September

Reading Fastdial's PBX records for the adherence report turned up the largest gap on the floor. It isn't how agents dial. It's what happens when a customer rings back.

The inbound gap

INBOUND, 14 DAYS 4,522 calls; 2 answered. All go to extension 716, which has no device, so the PBX sends them to voicemail.	CALLERS 2,960 1,752 of them are debtors in the book.	EVER CALLED BACK 27% of callers; the median wait for those who were was 46 hours.	WITHIN THE HOUR 2% A customer who rings a collections floor is the warmest call it gets all day.
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From FastDialCallBoxMaster.callcdr, 14 days to 27 September 2026, read-only. "Called back" means an outbound call to the number after the caller's last missed call. Inbound volume peaks 10:00–12:00 and 15:00–16:00, and continues to about 22:00.

Missed calls, and returning them first

`/desk/inbox` groups the PBX's missed inbound calls by caller and matches each caller to their account on the last nine digits of either phone column, because the PBX writes `07...`, `+2547...` and `2547...`. It shows who is waiting, for how long, the owning agent, and a Call back button. The dialler hands out waiting callers from the last two days before any campaign row, longest wait first, each to the agent who owns the account. When an agent's browser phone rings, the incoming screen now looks the caller up in the book, and answering opens their account.

The inbound line — the floor's own ACD

The media edge can register as 716. A call to 716 then rings the edge instead of reaching voicemail, and the edge asks ConnectDesk whom to ring. First comes the agent who owns the caller's account, if Ready, then the agents who have been Ready longest, each for a set number of seconds. A browser phone rings on the edge. A desk phone is rung back through the PBX over 716's own registration, so the ACD works before anyone uses WebRTC. If nobody is Ready, or ConnectDesk can't be reached, the edge answers busy and the PBX falls back to voicemail as today. Pausing a seat in the Command Centre stops it being rung. The only change on the Yeastar is 716's password, plus *when busy* → *voicemail* on 716.

Whoever's phone is rung sees who is calling. A browser phone shows the caller's account on its incoming screen. A desk-phone agent gets a card on their ConnectDesk screen with the customer's name, lender, balance and days late, plus *Open the account*. Once they pick up, it reads *On the line*, because the edge reports each pick-up the moment it happens. The header on every screen carries a red count of today's callers still waiting for a call back.

The dialler

The dialler hands customers out in the order a collections floor works them:

- customers who rang in and were not called back;
- the callbacks this agent booked in CallBox that are now due, with the note they left;
- this agent's promises falling due today and not yet paid;

- this agent's promises broken in the last three days with nothing paid: the day-after call;
- then the seat's campaign rows.

When a seat turns Ready with its dialler on, the agent's screen asks for the next customer. The server claims one (`DeskDialClaim`, unique per customer, day and attempt, so two agents are never handed the same person), opens the case file, and places the call through the normal phone path. Preview mode counts down 20 seconds with *Call now* and *Skip*. The rules live in one pure module and are tested:

- only on working days, inside the calling window;
- at most the day's attempts per customer, and never twice within 45 minutes;
- the seat's own campaign or bucket first, then the floor's priority order.

Nothing dials for a seat whose screen is closed. Predictive mode runs as progressive, because the Yeastar API rings the agent first. It is **off per floor** until a supervisor turns on *Hand out calls automatically*, so the first login on 1 October does not start calling customers by surprise.

The contact policy. Attempts per day stop the dialler hammering a number within one day. The contact policy looks across the last seven days and across every call the floor logged, including calls made from CallBox's own screens, because a customer rung fifteen times this week does not care which screen dialled them. Before a customer is handed out, the dialler checks their account's history in one read (13 ms for 60 accounts) and holds them back when any of these apply:

- they have had 10 calls this week, or 3 conversations;
- they paid in the last 2 days (campaign calls only);
- they promised to pay by a date that has not come yet (the reminder on the day still goes out);
- their latest outcome is one the supervisor listed as do-not-dial. The list is Fastdial's own; a *Suggest* button offers wrong or reallocated numbers, deceased, abroad, in court, disputed and cleared.

A customer who rang in is always called back, a callback booked with the customer ignores the weekly caps, and an agent can still call anyone by hand. On a sample of 60 accounts called in the last two days, 6 had already been spoken to four or five times that week. The numbers are set in Dialler settings → Contact policy.

Adherence

`/desk/adherence` measures each agent's day against the roster from the floor's own rows: first login and lateness, logged-in time, breaks by type and overruns, paused time, available time and occupancy. On a CallBox floor the PBX is a second witness. It gives dials, connects, talk time, first and last dial per extension, and dialling density in 15-minute slots on the day ribbon, so days worked on a desk phone show even before anyone signs in (marked *Desk phone*, never counted as present). A range view gives one dot per day and totals, and the CSV has one row per agent per day for payroll.

Recordings, and AI call review

30,372 of 30,525 answered outbound calls in 30 days carry a recording on the PBX. ConnectDesk now plays them in the case file's history and on the phone floor, using the same two API calls CallBox makes. A five-minute WAV is larger than a Vercel response may be, so audio is served in byte ranges of 4 MB or less, and the download assembles them. Supervisors can play any call and agents only their own. Every playback is audited.

A supervisor can also have one recording reviewed by the platform's AI model (Gemini 2.5 Flash, which hears English, Swahili and Sheng). The review gives a transcript in the language spoken, an English summary, and any promise to pay with the customer's own words. It adds the outcome the call resembles, drawn from Fastdial's own list, compliance flags with verbatim quotes (threats, abuse, the debt disclosed to someone else, misrepresentation, harassment), and a QA checklist. It is advice and never saves anything. The score is computed by ConnectDesk, not the model, and a threat or a disrespectful call caps it at 40. It is **off per organisation**, because the recording leaves Kenya for Google, and that is each lender's decision.

Staffing

`/desk/staffing` averages four weeks of the PBX's records per weekday and hour. For each hour it applies Erlang C to the inbound line at a chosen service level (for example, 80% answered within 20 seconds), with shrinkage, and compares the result with the rostered floor. On Fastdial's real week:

- The busiest inbound hour is Thursday 16:00, at 55 calls. Four agents on the line cover it, six once shrinkage is counted, which leaves about 25 of 31 free to dial.
- 10:00 is the best hour to dial: 28% of calls connect, against 26% across the day.
- Customers keep ringing until about 22:00, while the floor goes home at 17:30.

The wallboard, and quality review

`/desk/wallboard` is the floor on the TV. It is full screen and dark, with type sized for a room: today's calls, connects, promises, money in, and customers waiting for a call back. Every signed-in seat shows its state and timer from the live stream, and a break over its allowance turns red. A banner appears while customers are ringing in, alongside the promise leaderboard and the day hour by hour.

`/desk/quality` is the QA team's list. It covers recorded calls with the outcome the agent saved, for a day or a range, and one click turns it into a fresh random sample of N calls per agent. Filters cover agent, outcome and minimum talk time. Each call plays in its row, and AI scores and conduct flags roll up per agent where reviews exist. On the four days to 27 September there were 2,036 such calls across 24 agents, and a sample of five per agent gives 117 to listen to.

Client reports, and whether promises were kept

A call centre reports to each lender it works for. `/desk/reports` builds that report for one lender, or all of them, over any range up to 92 days. It covers accounts called and reached, promises taken, the promises falling due and what became of them, and the money collected. Each is broken down by day, by agent, by contract and by call outcome. The page prints light on A4, so *Save as PDF* produces the document, and each table downloads as CSV.

CallBox never records whether a promise was kept: `promisedtopay.amountpaid` is empty on every one of its 28,589 promises. ConnectDesk now judges each promise against the payments on the account. A promise is **kept** when payments from the day it was taken to 3 days after it fell due cover 95% of it. It is **part-paid** if something came in, **broken** if nothing did, and **open** until that window closes. The promise board and the case file carry the same verdict on every promise. Money is credited to the last agent who reached the account in the 30 days before it paid. The definitions are printed at the foot of every report.

CALLS, 21-27 SEP

31,820

to 17,124 accounts; 40% of them reached.

PROMISES TAKEN

4,529

worth KES 30.0M.

PROMISES KEPT

15%

312 of 2,067 whose window closed; 1,653 broken outright.

COLLECTED

4.57M

KES, 78% of it on accounts an agent had reached in the 30 days before.

All lenders, Fastdial's CallBox, read-only. For Fourth Generation Capital alone the kept rate was 12% by count and 3% by value. A promise that is never paid is the floor's largest leak, and until now it was invisible.

For the agent: My day, and a call guide

An agent opening ConnectDesk now sees their own day above the floor's. It shows their calls and accounts reached, promises taken, and their promises due today, with how much has been paid and how many are kept. It also lists their broken promises waiting for the day-after call, and the money credited to them this month. That credit uses the client report's own rule, so the agent's number and the one sent to the lender agree. The card also says where they stand on the floor today, by calls and by value promised.

Every case file carries a **call guide**. It sets out the seven things a good collections call does, in order, worded for the customer on screen:

- greet;
- say who you are and whom you call for;
- confirm it is them, and say nothing of the loan to anyone else;
- state what is owed;
- ask for payment, with the lender's paybill and account;
- agree a next step and read it back;
- stay respectful: never threaten arrest, seizure or shaming, and never pose as police, a court, CRB or a lawyer.

It is the same checklist the AI review scores recordings against, so an agent is guided by exactly what they are judged on. Uploaded books carry batch tags and numbers in the name column (for example "... KCB CONV LIVE SME'S" or "... REPEATER"), so the guide gives a name that can be said aloud. It also names the lender as a customer knows it.

Each campaign card now shows what the campaign has produced since it opened: accounts reached, promises taken, and money paid on the accounts it called. A supervisor can see which slices of a book are worth the floor's time.

Proven how

Five tests run with no database: `npm run test:dialler` (47 checks on a UTC clock, 20 of them the contact policy and the broken-promise call), `test:report` (44: promise verdicts at every boundary, ranges, rates, CSV escaping, speakable names), `test:edge` (27: the inbound line's registration and dialplan), `test:call-review` (27: what may reach the screen from the model) and `test:staffing` (19: Erlang C against the closed form). Every page was rendered on Fastdial's real data with no console errors. The telephony bridge's new file route was tested against a local fake PBX, byte for byte. Two things could not be run from this workstation: a recording download from the PBX (it reads a customer's call) and the AI review itself (the model key lives on Vercel). Both run on the founder's first click in production.

08 Go-live — Fastdial

#	STEP	WHO	NOTES
1	Sign in at <code>connectdesk.servicesuitecloud.com/fastdial</code> as Fastdial's admin, then Settings → Connections → Test the PBX from this server	You	"Logged in. 50 extensions..." means Vercel can drive the PBX. Error 20015 (IP refused) means step 2.
2	Only if step 1 said 20015: run the telephony bridge on <code>servicesuite-in (bridge/telephony-bridge)</code> , set its URL and secret in the phone settings, and add 213.148.17.228 to the Yeastar API's allowed IPs	You	≈20 min. The bridge also runs the floor's minute scheduler, which is worth having either way.
3	Settings → Seats → Bring agents in from CallBox : select the agents and hand each their one-time password	You / Fastdial	Each gets a login (Agent role), a seat and their extension from CallBox. There is no SMS sender on Fastdial yet, so sign-in is password only. It is audited, and adding an SMS sender turns on the daily code.
4	An agent signs in, opens a campaign, presses Call : their desk phone rings, then the customer. Save the outcome	You at Fastdial	This is the day-one proof. Check that CallBox shows the same outcome on the same row.
5	Supervisor: Command centre → start the floor → a seat → Listen	You	Needs the supervisor's own extension on their seat and monitor permissions on the Yeastar (edge runbook, "On the Yeastar").
6	Browser phone: install the edge on a small Ubuntu box on Fastdial's LAN (<code>bridge/edge/install.sh</code>), set its address and each seat's SIP password in ConnectDesk	You	≈45 min plus DNS. Log MicroSIP or Linkus out on those PCs first, because an extension holds one registration.
7	Point <code>CRON_SECRET</code> for the bridge (or any 1-minute scheduler) at <code>/api/cron/desk-floor</code>	You	The live stream ticks the floor while screens are open; this covers the rest.
8	Inbound line : in Settings → Phone system enter 716 and its registration password; on the Yeastar set 716's <i>When busy</i> forwarding to Voicemail	You	Needs the edge (step 6). Within a minute <code>pjsip show registrations</code> shows <code>reg-716 Registered</code> . Ring 716 from a mobile: a Ready agent's phone rings.
9	Dialler settings → Hand out calls automatically , once agents have made a few calls by hand	You / Fastdial's supervisor	Start with <i>preview</i> mode on a few seats (a 20-second look before each call), then <i>progressive</i> . Missed callers are handed out first. First check Contact policy on the same page and press <i>Suggest</i> for the do-not-dial outcomes.
10	Open a case file with a recorded call and press play	You	Proves the recording path from Vercel. If it fails with 20015, the bridge from step 2 carries recordings too.

11	Only once Fastdial has agreed it in writing: Settings → Phone system → AI call review , then <i>Review</i> on one recording	You / Fastdial	The first review proves the model key on Vercel. Customers must be told calls are recorded and reviewed.
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09 Go-live — Micromart

The live floor is already Micromart's: 26 CollectBox agents seated, the seven bands with real counts, supervisors on the Command Centre. What calling needs from Micromart:

- **What the PBX is** (make, model, firmware) and its admin/API access. If it is a Yeastar S-Series, the driver exists. If it is Asterisk/FreePBX or 3CX, a driver slots in behind the same **PhoneSystem** interface, and the edge works with any SIP PBX regardless.
- **Each agent's extension and its SIP password**, entered on the seats.
- Their CollectBox **calldr** stopped being written in September 2023, so ConnectDesk's own call ledger (DeskCall) will be the call record until their PBX's events are connected.

10 Risks, and what is not done

ITEM		RECOMMENDATION
CallBox is connected as the Postgres superuser (the same credential sits in plain text in CallBox's appsettings.json, for a port that answers from the internet)	Risk	Create a connectdesk login with rights on the CallBox tables only, and replace the string in Settings → Connections (tested before it saves). Rotate the superuser password and restrict port 2020.
Writes to CallBox are on for Fastdial	Decision	Needed for the pilot. One switch in Settings → Connections makes the floor read-only.
The PBX is full: 50 of 50 extensions	Decision	Reclaim the idle ones before adding anyone.
Recording and transcription	Legal	Customers must be told calls are recorded; recordings are personal data under the Data Protection Act 2019. Agree retention and access with both companies before the edge records anything. Playback in ConnectDesk is limited (agents hear only their own calls) and audited.
AI call review sends a customer's voice to Google	Legal	A cross-border transfer of personal data. Off per organisation; turn it on only with the company's written agreement and a notice to customers. Every review is logged.
Nobody answers 716 after hours	Decision	Customers ring until about 22:00; the floor stops at 17:30. The inbound line falls back to voicemail and the missed-calls inbox, but an evening shift of two covers most of it (Staffing page).
Not built yet	Next	True predictive dialling (dialling ahead of agents through the edge's own queue). Live transcription and co-pilot during a call (AI review is after the call). A driver for Micromart's PBX once it is known.

11 Where it lives

PATH (CONNECTED-SUITE)	ROLE
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<code>lib/collectbox/tenant.ts</code>	The tenant gate on every CollectBox read
<code>components/onboard/*</code> , <code>app/api/orgs/*</code> , <code>lib/suite/links.ts</code>	Onboarding v2 and every system's address
<code>lib/desk/source.ts</code> , <code>lib/desk/nav.ts</code>	Which floor an org has, and its menu
<code>lib/callbox/{client,floor,write}.ts</code> , <code>components/desk/callbox/*</code>	CallBox: connection, reads, CallBox-exact writes, screens
<code>lib/desk/floor/server.ts</code> , <code>app/api/desk/floor/*</code>	Phase 2: the floor on the server, the stream, the scheduler
<code>lib/desk/telephony/*</code> , <code>lib/desk/calls.ts</code>	The PhoneSystem interface, the Yeastar S driver, the call pipeline
<code>components/desk/phone/{useCallEngine,RealCall}.tsx</code>	The phone's engine: desk phone, WebRTC, the in-tab conference
<code>lib/desk/{settings,seats,edge}.ts</code> , <code>components/desk/settings/*</code>	Connections, seats and agent import, the generated edge config
<code>bridge/edge/*</code> , <code>bridge/telephony-bridge/*</code>	The media edge kit and runbook; the fixed-IP bridge
<code>lib/desk/adherence.ts</code> , <code>app/desk/adherence</code>	The adherence report and its CSV
<code>lib/callbox/inbound.ts</code> , <code>app/desk/inbox</code>	Missed calls, callback status, caller lookup
<code>lib/desk/dialler/{engine,rules}.ts</code> , <code>components/desk/dialler/AutoDialler.tsx</code>	The dialler: claims, rules, the agent's screen
<code>lib/desk/inbound-route.ts</code> , <code>app/api/desk/edge/inbound</code>	The inbound line's hunt list (the ACD)
<code>lib/desk/recordings.ts</code> , <code>lib/desk/ai/call-review.ts</code>	Recording access, and the AI review with its safety layer
<code>lib/desk/erlang.ts</code> , <code>lib/callbox/profile.ts</code> , <code>app/desk/staffing</code>	Erlang C and the week's volume profile
<code>lib/desk/wallboard.ts</code> , <code>components/desk/Wallboard.tsx</code>	The wallboard
<code>lib/callbox/quality.ts</code> , <code>app/desk/quality</code>	Quality: recorded calls, sampled per agent
<code>components/desk/phone/InboundPop.tsx</code> , <code>app/api/desk/floor/stream</code>	The desk-phone screen pop, fed by the stream's <code>ring</code> event
<code>lib/callbox/report.ts</code> , <code>lib/desk/report-{rules,csv}.ts</code> , <code>app/desk/reports</code>	Client reports, the promise verdict, the CSV
<code>lib/callbox/contact.ts</code> , <code>lib/desk/dialler/rules.ts</code> (<code>contactBlock</code>)	The contact policy: the week's history, and who is held back
<code>lib/callbox/agent-day.ts</code> , <code>components/desk/callbox/CallGuide.tsx</code> , <code>lib/desk/{spoken,ai/qa-checks}.ts</code>	The agent's My day; the call guide, its checklist and speakable names
<code>scripts/verify-{dialler,report,edge,call-review,staffing}.ts</code>	The five tests (<code>npm run test:...</code>)
<code>sql/connectdesk.sql</code>	The schema delta (applied): Org profile, vault kinds, the Desk tables including <code>DeskDialClaim</code> and <code>DeskCallReview</code>

ConnectDesk · Pilot build · 28 September 2026 · commits 6ba07b0, 43f789a, 94fe1c7, bd5310a, 66b3dbf, 19f8f9c, 098707a, 1c5ff25, 7e1a4ee, 95a5b0b, a11eb17, e8e6a4d, bbceba3, 1b66703, 4a04435, 00b3184, aa01f30, a16af07, c3c510f, b18da20, 78603c3, b4926ba on connected-suite main, each deployed to production. All CallBox writes were proven in rolled-back transactions. From this workstation the PBX was contacted only for a login and an extension list; its call records were read, never written.